

PharmaSafe OCR

Automated Prescription Information Extraction

Three-stage deep learning pipeline

ROI Detection

Handwritten OCR

Drug Correction

PharmaSafe Full Pipeline — Presc1.jpg.rf.31b6b89e77031a5bcb9c04bd277b237d.jpg
(? = low-confidence spell correction, verify manually)

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Name: Bahar Bhatti Age: Gender: Date: 01/01/2023
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Weight:	15 kg (64%)
BP: 100/70 mmHg	100/70 mmHg
Pulse:	100 bpm (64%)
Temp:	37.5°C
T.D.M	name: Se dose: Lumox
DCM	Tab 100 mg
BCP	100 mg

From raw prescription image → structured medication fields

The healthcare problem

Handwritten prescriptions are high-risk, inconsistent, and hard to standardize.

Why automation matters

- Drug names and doses are often handwritten in non-standard formats
- Medical abbreviations and mixed languages increase ambiguity
- Errors can propagate between physicians, pharmacists, and nurses
- Structured extraction enables safer pharmacy processing and downstream analytics



PharmaSafe takes a middle path

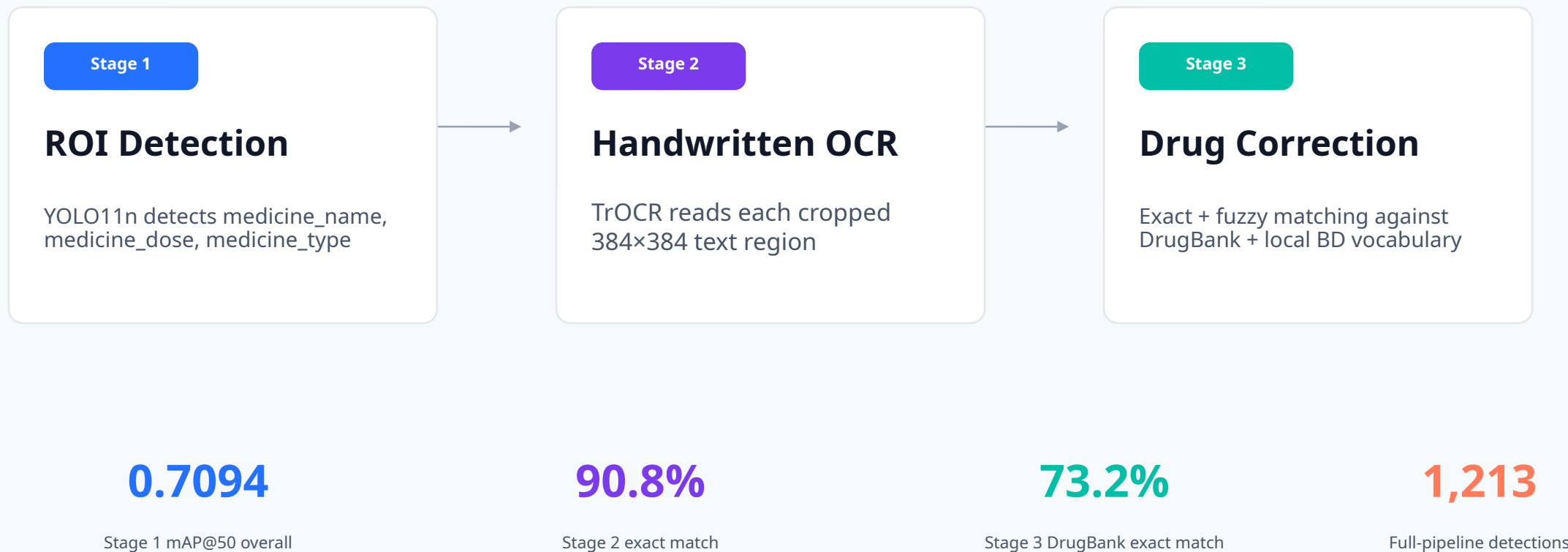
Localize only relevant fields, recognize each crop with specialized handwriting OCR, then validate drug names against pharmaceutical vocabulary.



Traceable, modular, safer

Pipeline architecture

A modular design: each stage can be trained, evaluated, and improved independently.



Stage 1 — ROI Detection with YOLO11n

Detecting the medication fields before OCR makes the pipeline more precise and interpretable.

1

ROI Detection

- Target classes: medicine_name, medicine_dose, medicine_type
- Merged 3 Roboflow prescription datasets from South Asia
- Class labels remapped to a unified 3-class schema
- Stratified 70 / 15 / 15 train-validation-test split

Why YOLO11n?

- Fast single-stage detection for real-time pipeline use
- Nano model limits overfitting on ~1,200 images
- COCO pretraining + frozen backbone reduces data needs
- Matched larger YOLO variants on validation at this scale

300 epochs max

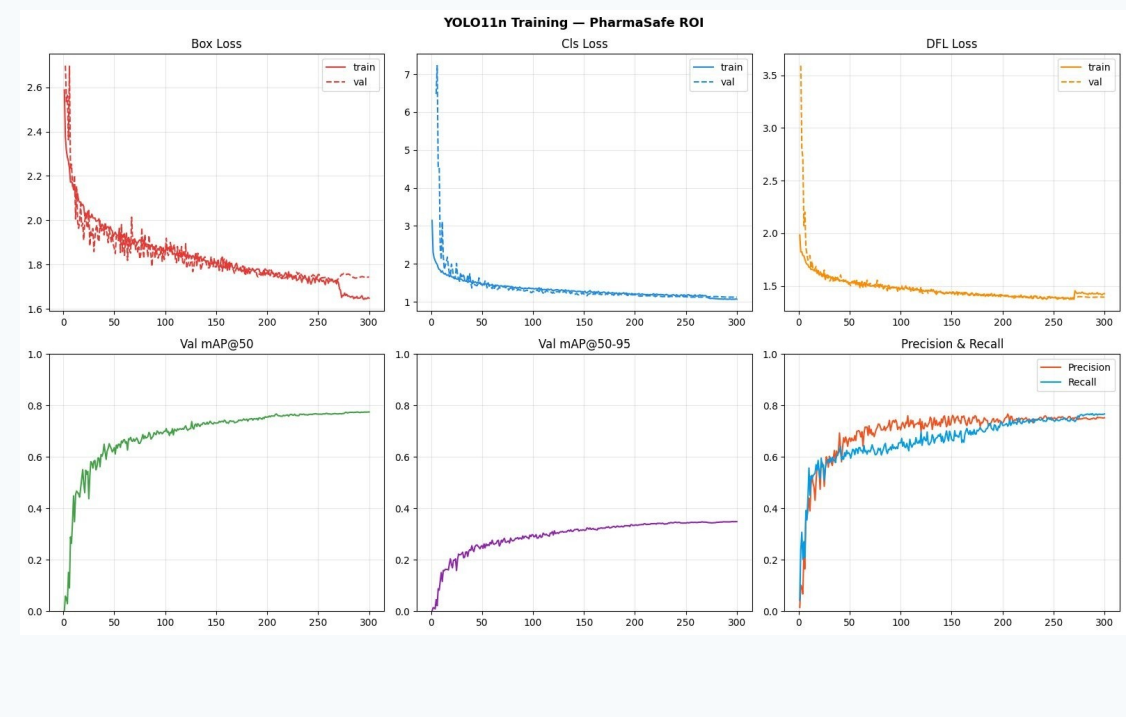
AdamW

640×640

mosaic aug

Stage 1 results

Strong region detection with clean training convergence and interpretable visual outputs.



0.7840

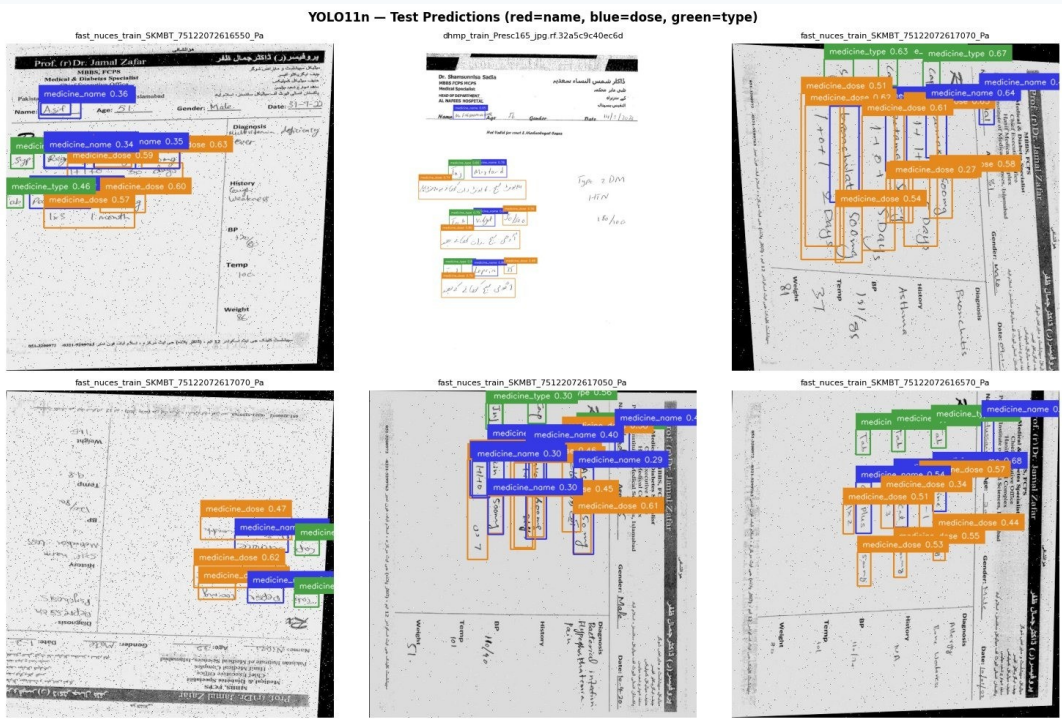
Mean F1

0.7959

Mean Recall

0.7094

mAP@50 overall



0.3227

mAP@50-95

Interpretation: the detector reliably finds relevant regions at IoU@50, while tighter localization remains harder because prescription boxes are loosely annotated.

Stage 2 — Handwritten OCR with TrOCR

Each detected crop becomes a word-level OCR task.

2

Handwritten OCR

- Input: 384×384 RGB crop from Stage 1
- Model: microsoft/trocr-base-handwritten
- Datasets: RxHandBD + Kaggle BD handwritten prescription data
- Training strategy: decoder-only warmup, then full-model fine-tuning

Why TrOCR?

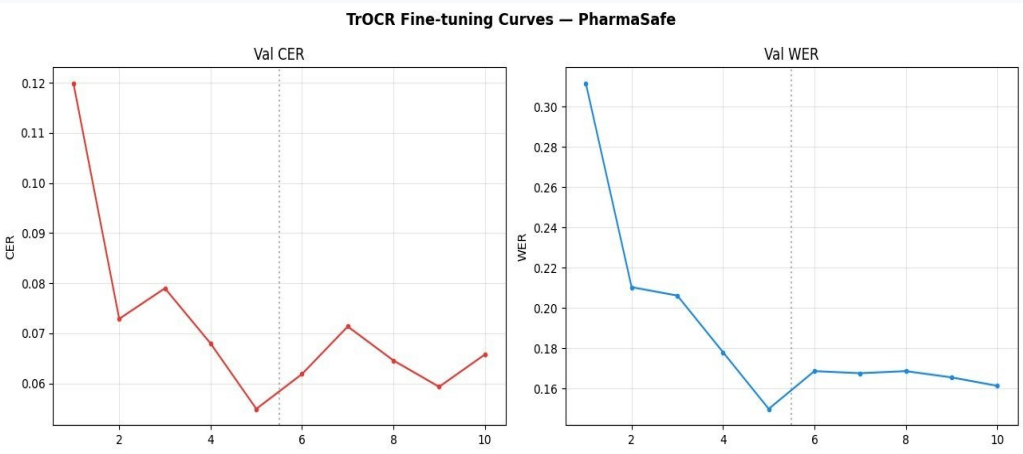
- Designed for handwriting, unlike traditional OCR
- Transformer encoder-decoder handles variable-length text
- Better contextual modeling than CRNN-style OCR
- Domain fine-tuning learns drug-name sequences



crop → text prediction

Stage 2 results

High exact-match performance on handwritten drug names.



90.8%

Exact Match (830 / 914)

0.0647

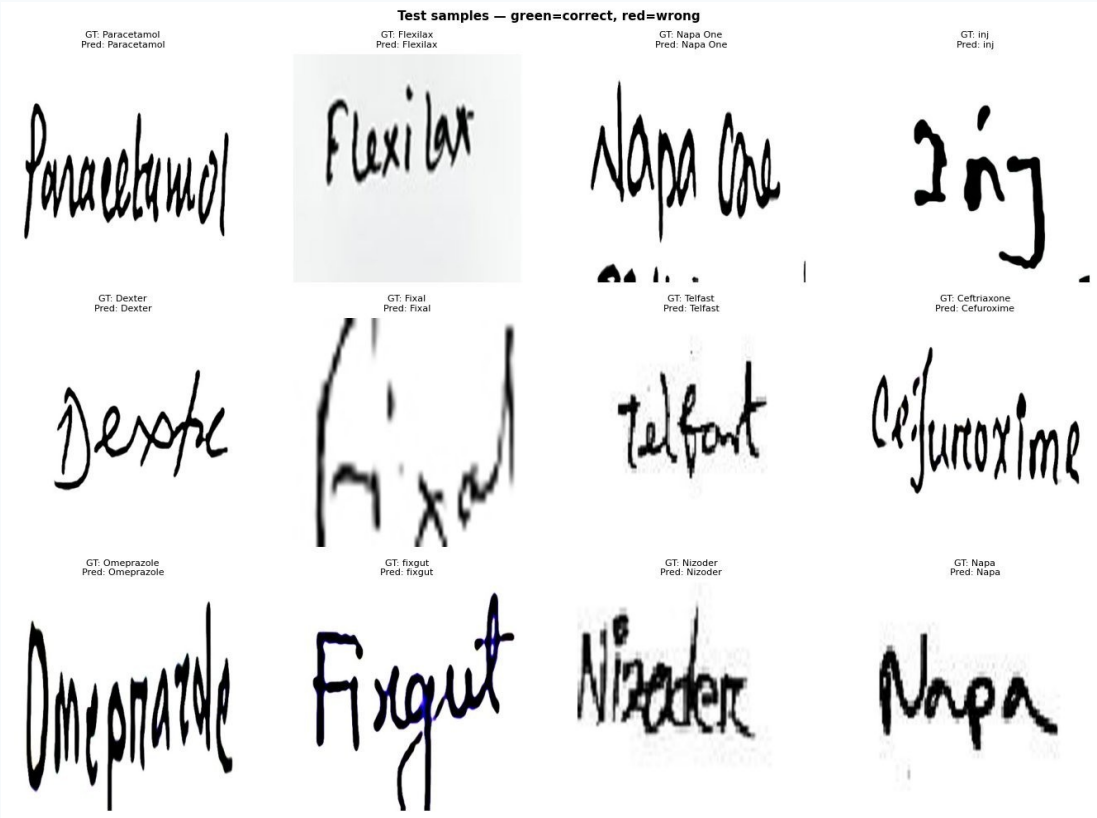
Character Error Rate

0.161

Word Error Rate

~0.91

RapidFuzz Drug F1 ≥ 90



Main error pattern: visually similar drug names, such as Ceftriaxone vs Cefuroxime.

Known limitation — dosage OCR

Dose fields behave differently from drug-name fields.

1

Pre-training bias

TrOCR was trained mostly on words, so digits can become letters: 500mg → SOO mg.

2

Data imbalance

Fine-tuning data mostly contains medicine-name crops, with fewer dose-number examples.

3

Notation complexity

Dose fields often use BD, TDS, SOS, or 1+0+1; spell correction is not applied to dose fields.

Future fix: add dose-specific training data + numeric post-processing rules.

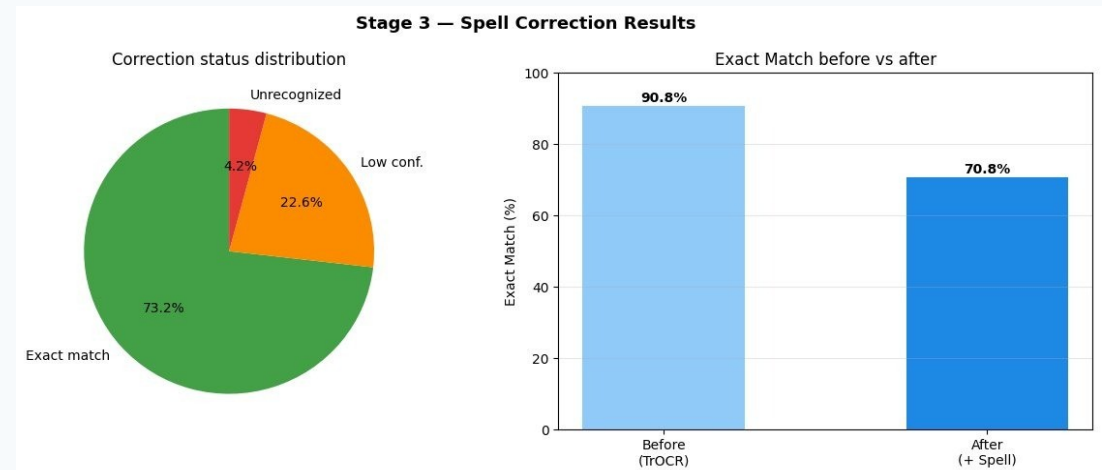
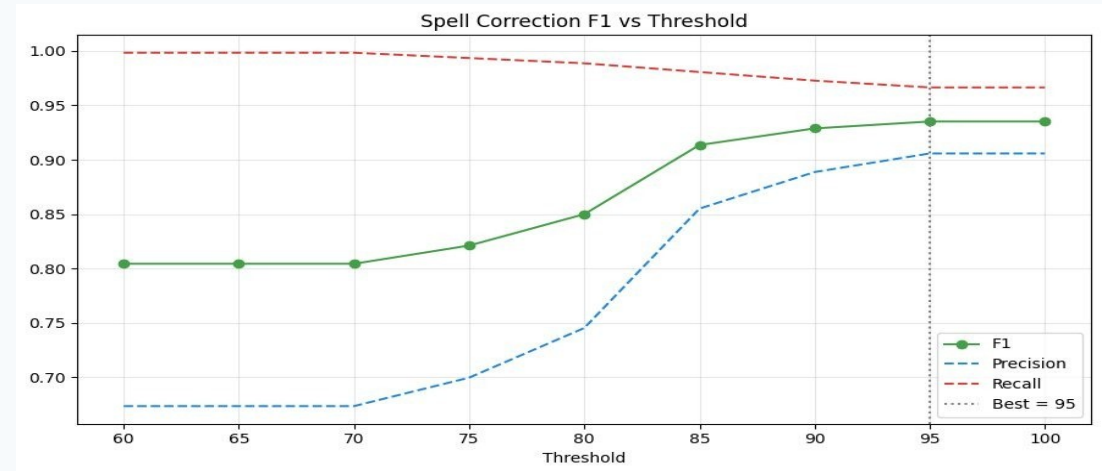
Stage 3 — Drug name spell correction

OCR text is standardized using pharmaceutical vocabulary instead of a general dictionary.

3

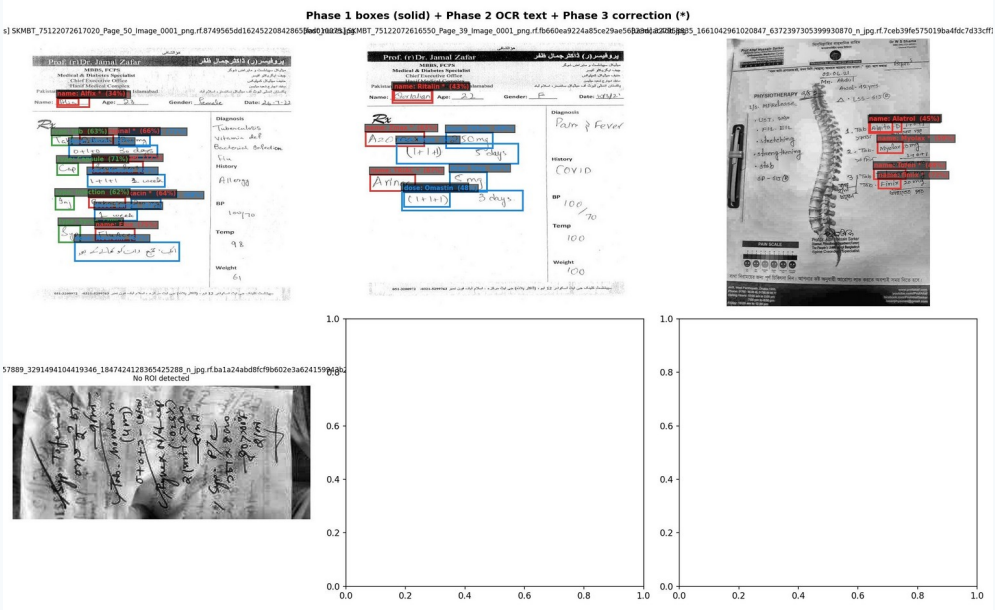
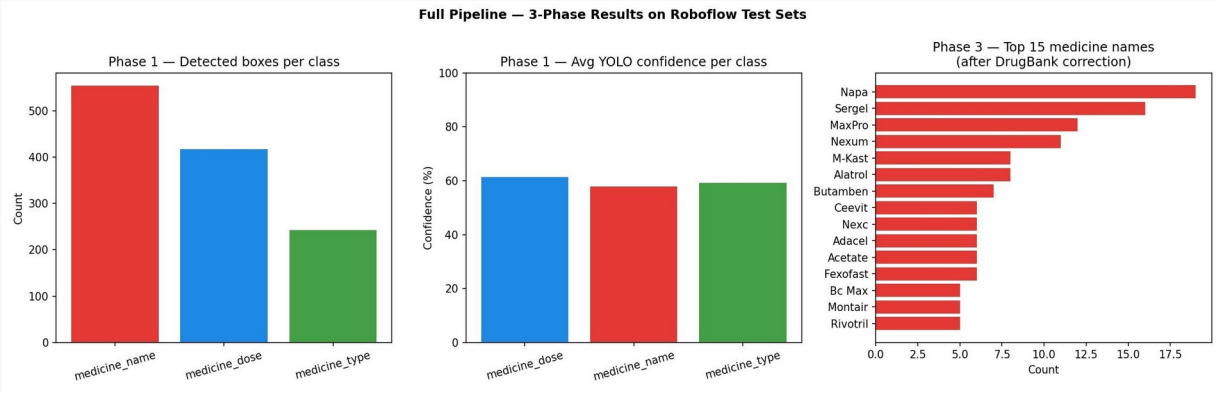
Drug Correction

- Vocabulary: DrugBank generics + synonyms + brand names, plus local BD labels
- Pass 1: exact case-insensitive lookup
- Pass 2: RapidFuzz token_sort_ratio with cutoff 75
- High-confidence threshold selected at 95 using F1 sweep



Full pipeline integration

All stages are connected into one extraction flow and exported to pipeline_results.csv.



111

Images processed

106

Images with detections

1,213

Total crops/detections

Class split: medicine_name 554 • medicine_dose 417 • medicine_type 242

Single prescription demo

The same pipeline was tested on an unseen prescription image.

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Name: Behas Bibi Age: Gender: Date: 24/01/2023

NOT VALID FOR MEDICOLEGAL & COURT CASES

Weight: BP: 100/70 mmHg Pulse: Temp: T-LDM DCM BCP

Tab Aldosterone 120mg
Tablet (64%) Se: Flamyd
Tab Clonidine
name: Se dose: Lumox
Tab

What the demo shows

- YOLO detects medicine_name, medicine_dose, and medicine_type boxes
- TrOCR reads each region independently
- DrugBank correction validates recognized medicine names
- Low-confidence corrections are marked for manual pharmacist review

Safety note: "?" flags should not be auto-accepted in a medical setting.

Limitations & future work

The core bottleneck is not architecture — it is publicly available data quality.

Current limitations

- No unified public dataset with boxes + OCR + standardized drug names
- Cross-dataset domain gap between detection and OCR sources
- Dosage OCR is weaker due to numeric/abbreviation complexity
- English drug-name coverage only; Bengali/Urdu/Hindi remains limited

Future improvements

- Build a unified public prescription dataset
- Explore end-to-end training once integrated labels exist
- Upgrade to larger YOLO variants with more data
- Add multilingual OCR and drug-name language models
- Deploy active learning with pharmacist feedback

Final takeaway

PharmaSafe proves that a transparent, modular deep-learning pipeline can extract medication information from handwritten prescriptions using fully public datasets.

Detect

find medicine fields

Read

recognize handwriting

Correct

standardize drug names

Questions?